

1941-1952

THE NUCLEAR ARMS RACE

Nuclear weapons technology was first developed in 1941 under the US Manhattan Project. This resulted in the creation of a fission weapon which was tested in New Mexico and used to devastating effect on the Japanese cities of Hiroshima and Nagasaki in August 1945. US authorities kept the nuclear programme secret from the USSR, its wartime ally, but information leaked to the Soviets via a network of spies. The Soviets worked clandestinely to develop their own bomb, surprising Western experts by successfully testing it before the end of the decade. Meanwhile

other nations were also working on nuclear projects of their own: Britain conducted its first nuclear test in 1952, France in 1960, and China in 1964.

The two superpowers raced to upgrade their nuclear arsenals through the development of far more powerful fusion weapons. US scientists led the way, exploding a thermonuclear device at Enewetak Atoll in 1952, but the Soviets were not far behind, testing a less sophisticated device just nine months later. Their rivalry made the world a much more dangerous place for future generations.

KEY MOMENTS

29 August 1949 First Soviet atomic test

US authorities were taken by surprise when a plane conducting a weather reconnaissance flight in the northern Pacific detected traces of radioactivity in the atmosphere off Russia's Kamchatka Peninsula. To their alarm, their Cold War opponents had successfully tested a bomb, dubbed "First Lightning" (*left*). The Soviets devoted vast resources – including the use of forced labour from the Gulags – to mine the uranium needed to produce their weapons.

January 1950 Soviet spy arrested

One reason for the Soviet success, besides the discovery of new uranium sources in eastern Europe, was technical information passed clandestinely by Klaus Fuchs (*left*), a German physicist who carried out nuclear research in the US from 1944 to 1946 and then in Britain. In secret a committed Communist, Fuchs confessed and received a 14-year jail sentence.

